

1. 請框出答案. 2. 不可使用手機、計算器，禁止作弊!

1. For the following, mark True or False. Justify your answer with a graph if true, or with a counterexample if false. (若為真，提供圖例顯示；若為否，提供反例)

True False The span of any two nonzero vectors in $\mathbb{R}^2$ is all of $\mathbb{R}^2$.

Solution :

Similar with 1-1 problem 39, (f)(g).

2. Find c such that $\vec{u} = [c, c + 2, c - 1]$ is parallel to the vector $\vec{v} = [9, 15, 6]$.

Answer: $c =$ 3

Solution :

$$\begin{cases} c = 9r \\ c + 2 = 15r \\ c - 1 = 6r \end{cases} \Rightarrow r = 1/3, c = 3$$

3. Let $\vec{v}$ and $\vec{w}$ are any two vectors in $\mathbb{R}^n$, and let r be any scalar in $\mathbb{R}$. Please prove the following property.

$$r(\vec{v} + \vec{w}) = r\vec{w} + r\vec{v}.$$

Solution :

Similar with example 4 from 1-1. Notice that the order of $\vec{v}$ and $\vec{w}$ is not the same on both sides of the equation.

Let $\vec{v} = [v_1, v_2, \dots, v_n]$ and $\vec{w} = [w_1, w_2, \dots, w_n]$.

$$\begin{aligned}
 LHS &= r(\vec{v} + \vec{w}) \\
 &= r([v_1, v_2, \dots, v_n] + [w_1, w_2, \dots, w_n]) \\
 &= r[v_1 + w_1, v_2 + w_2, \dots, v_n + w_n] && \text{by Definition of Vector addition} \\
 &= [r(v_1 + w_1), r(v_2 + w_2), \dots, r(v_n + w_n)] && \text{by Definition of Scalar multiplication} \\
 &= [rv_1 + rw_1, rv_2 + rw_2, \dots, rv_n + rw_n] \\
 &= [rv_1, rv_2, \dots, rv_n] + [rw_1, rw_2, \dots, rw_n] && \text{by Definition of Vector addition} \\
 &= r[v_1, v_2, \dots, v_n] + r[w_1, w_2, \dots, w_n] && \text{by Definition of Scalar multiplication} \\
 &= r\vec{v} + r\vec{w} \\
 &= r\vec{w} + r\vec{v} = RHS && \text{by A2}
 \end{aligned}$$

Definition 1.1: Vector Algebra in $\mathbb{R}^n$

Let $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{w} = [w_1, w_2, \dots, w_n]$ be vectors in $\mathbb{R}^n$. Let r is any scalar. We define the following:

Vector addition/subtraction: $\mathbf{v} \pm \mathbf{w} = [v_1 \pm w_1, v_2 \pm w_2, \dots, v_n \pm w_n]$

Scalar multiplication: $r\mathbf{v} = [rv_1, rv_2, \dots, rv_n]$

Theorem 1.1: Properties of Vector Algebra in $\mathbb{R}^n$

Let $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ be any vectors in $\mathbb{R}^n$, and let r and s be any scalars in $\mathbb{R}$.

Properties of Vector Addition

A1: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$

A3: $\mathbf{0} + \mathbf{v} = \mathbf{v}$

A2: $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$

A4: $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$

Properties Involving Scalar Multiplication

S1: $r(\mathbf{v} + \mathbf{w}) = r\mathbf{v} + r\mathbf{w}$

S3: $r(s\mathbf{v}) = (rs)\mathbf{v}$

S2: $(r + s)\mathbf{v} = r\mathbf{v} + s\mathbf{v}$

S4: $1\mathbf{v} = \mathbf{v}$